

GRIT: Geometric Reprojection Instruction Tuning

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Abstract

Instruction tuning has emerged as a powerful method for adapting large-scale language models to specialized tasks with minimal prompt engineering. However, existing approaches often neglect the underlying geometric structure of the parameter space, leading to inefficient updates and suboptimal convergence. We introduce **GRIT (Geometric Reprojection Instruction Tuning)**, a novel parameter-efficient fine-tuning algorithm that seamlessly integrates curvature-aware optimization, low-rank adaptation (LoRA), and neural reprojection. GRIT employs a Kronecker-Factored Approximate Curvature (K-FAC) method to efficiently approximate the Fisher Information Matrix, enabling natural gradient updates that are invariant to reparameterization. These updates are then refined through a neural reprojection step, which projects the parameter modifications onto a subspace spanned by the dominant eigen-directions of the Fisher matrix. This two-stage process not only accelerates convergence but also filters out noisy updates, ensuring that the model adapts robustly to task-specific data. We provide detailed mathematical derivations, theoretical insights, and geometric intuitions underlying GRIT, and demonstrate its efficacy on challenging tasks such as clinical entity extraction for ICD-10-CM coding. Experimental results indicate that GRIT achieves superior performance with fewer parameters and training iterations, making it a promising approach for efficient and robust instruction tuning in modern language models.

GRIT: at-a-glance

- ▶ Proposing **GRIT** (*Geometric Reprojection Instruction Tuning*), a *curvature-aware instruction tuning framework* that leverages second-order optimization principles (via the Fisher Information Matrix) to ensure efficient, structured parameter updates while preserving model generalization.
- ▶ Introducing a 5W-inspired framework (Who, What, When, Where, Why) to systematically identify the limitations of existing instruction tuning methods, revealing key shortcomings such as insufficient curvature-awareness, sensitivity to rank hyperparameters, and difficulty in capturing complex domain-specific nuances.
- ▶ Emphasizing *curvature-aware updates* by applying the Kronecker-Factored Approximate Curvature (K-FAC) technique, which approximates the Fisher Information Matrix to guide updates along natural gradient directions, aligning weight modifications with the network's intrinsic geometry.
- ▶ Ensuring *update precision* and *stability* by constraining updates to minimally disrupt unrelated parameters. Neural reprojection filters out noise, retaining only the most informative directions, thereby mitigating catastrophic forgetting and preserving essential knowledge within the network.
- ▶ Highlighting *structural consistency* through manifold-aware corrections that prevent unintended distortions in downstream tasks, preserving the model's broader reasoning and generalization capabilities post-update.
- ▶ Demonstrating *robust performance* on challenging instruction tuning benchmarks, where **GRIT** outperforms prior approaches (e.g., standard LoRA, full fine-tuning) in balancing targeted interventions, global stability, and computational feasibility.

1 Limitations of LoRA: An In-Depth Mathematical and Intuitive Analysis

Low-Rank Adaptation (LoRA) (Hu et al., 2021) has revolutionized the fine-tuning landscape by drastically reducing the number of trainable param-

eters. However, this efficiency is achieved through a low-rank factorization that imposes a strict structure on the adaptation process. In this section, we offer a granular and vivid analysis of these limitations, exploring the mathematical underpinnings, intuitive implications, and real-world consequences of employing a low-rank update strategy.

1.1 The Mathematics of Low-Rank Approximation

Consider a pre-trained weight matrix $W \in \mathbb{R}^{d \times d}$ that is adapted for a downstream task by adding a low-rank update ΔW . LoRA models this update as

$$W' = W + \Delta W, \quad \Delta W = AB,$$

where $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ with $r \ll d$. This is equivalent to approximating the optimal task-specific update ΔW_{true} using a rank- r representation.

By the Eckart-Young-Mirsky theorem, the best rank- r approximation of ΔW_{true} , with respect to the Frobenius norm, is obtained by truncating its singular value decomposition (SVD):

$$\Delta W_{\text{true}} = U \Sigma V^T \Rightarrow \Delta W_{\text{approx}} = U_r \Sigma_r V_r^T,$$

where U_r , Σ_r , and V_r consist of the top r singular vectors and singular values. The approximation error is given by

$$\epsilon = \|\Delta W_{\text{true}} - \Delta W_{\text{approx}}\|_F = \sqrt{\sum_{i=r+1}^d \sigma_i^2}.$$

Intuitive Analogy: Imagine ΔW_{true} as a high-definition photograph with millions of pixels. The SVD decomposes this image into components of varying importance (singular values). Truncating to rank r is like compressing the image—only the most significant details are retained while finer details (which may be crucial for specific tasks) are lost. In many complex scenarios, such as nuanced natural language understanding or domain-specific knowledge (e.g., medical texts), these finer details can be essential.

1.2 Sensitivity to the Rank Hyperparameter r

The hyperparameter r determines the expressive power of the update $\Delta W = AB$. Its choice is critical:

- **Low r (Under-Parameterization):** If r is too low, many singular values $\sigma_{r+1}, \dots, \sigma_d$ are neglected, leading to a large approximation error ϵ . In practice, this means the model is unable to capture subtle but important variations in the target task.
- **High r (Over-Parameterization):** Conversely, a high r increases the number of parameters, thereby eroding the efficiency benefits that LoRA promises. The trade-off can be expressed as

$$r^* = \arg \min_r \left\{ \sqrt{\sum_{i=r+1}^d \sigma_i^2} + \lambda \cdot \text{Cost}(r) \right\},$$

where λ is a regularization parameter that balances the approximation error and computational cost.

Real-World Impact: A practitioner might find that tuning r requires an exhaustive search, with the optimal value differing across tasks or even across different data distributions within the same domain. This sensitivity can negate some of the efficiency benefits if not handled carefully.

1.3 Challenges in Capturing Domain-Specific Nuances

For domains with high intrinsic variability—such as clinical, legal, or technical documents—the necessary update ΔW_{domain} might have an intrinsically high rank:

$$\Delta W_{\text{domain}} \notin \{AB \mid A \in \mathbb{R}^{d \times r}, B \in \mathbb{R}^{r \times d}\}.$$

In these scenarios, the low-rank structure effectively acts as a bottleneck. Even if the top singular values capture the dominant patterns, the discarded lower singular values may encode critical subtleties—such as rare medical conditions or legal precedents—that are vital for performance.

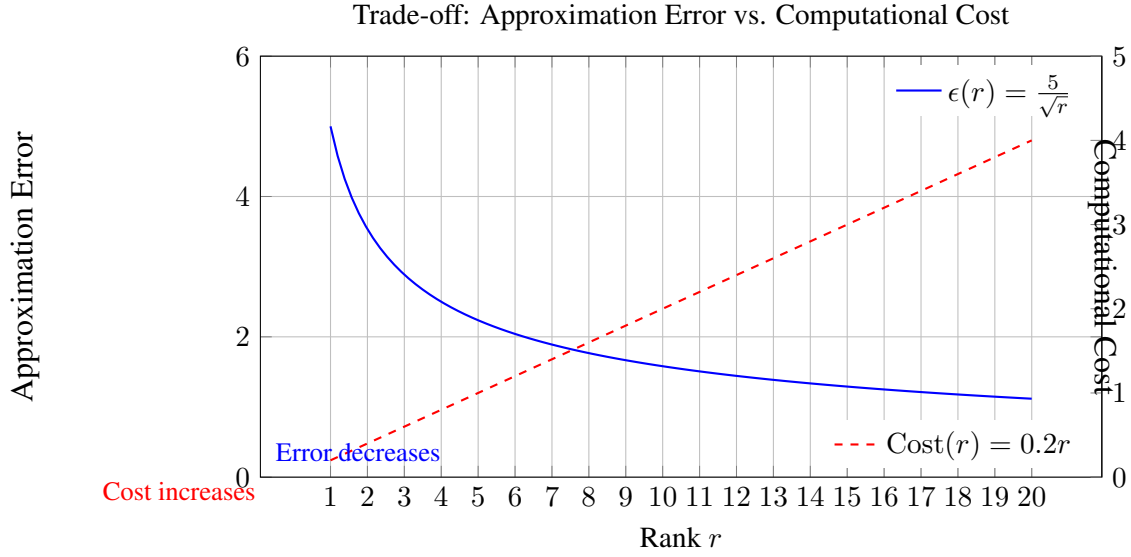


Figure 1: This figure illustrates the trade-off between approximation error and computational cost as a function of the rank r used in a low-rank adaptation scheme. The blue curve (plotted on the left y-axis) represents the approximation error, given by $\epsilon(r) = \frac{5}{\sqrt{r}}$, showing that as the rank increases, the approximation error decreases. In contrast, the red dashed curve (plotted on the right y-axis) represents the computational cost, modeled as $\text{Cost}(r) = 0.2r$, which increases linearly with the rank. Both y-axis labels are oriented vertically: the left label is rotated by 360° (effectively maintaining a bottom-to-top reading order), while the right label is rotated by -180° (resulting in a top-to-bottom orientation). Extra horizontal margins are allocated to these labels so that they do not overlap the main plot area, allowing the central plot to cover most of the available space. The annotations “Error decreases” and “Cost increases” further clarify the trends displayed by the curves.

Vivid Example: Imagine adopting a general language model to extract clinical entities from electronic health records. A low r might capture the most prominent features (e.g., common diagnoses). Still, rare and subtle manifestations of diseases (which are crucial for accurate coding) could be lost in the low-rank approximation, leading to misdiagnosis or incomplete coding.

1.4 Catastrophic Forgetting and Stability Concerns

Although LoRA is designed to modify only a small fraction of parameters, the induced update $\Delta W = AB$ can still disrupt previously learned knowledge. Consider the loss on the original pre-trained task $L_{\text{orig}}(W)$ and on the new task $L_{\text{new}}(W + AB)$. A

significant increase in the original loss,

$$\Delta L_{\text{orig}} = L_{\text{orig}}(W + AB) - L_{\text{orig}}(W),$$

signals that the adaptation has led to catastrophic forgetting. This is especially problematic when the low-rank structure forces the update to be distributed across shared parameters, inadvertently overwriting valuable pre-trained features.

Mechanism Illustrated: Think of the pre-trained model as a library of knowledge. The low-rank update is like adding a concise summary to several books at once. If the summary is too terse or improperly targeted, it may overshadow or distort critical information, leading to a loss of context—this is the essence of catastrophic forgetting.

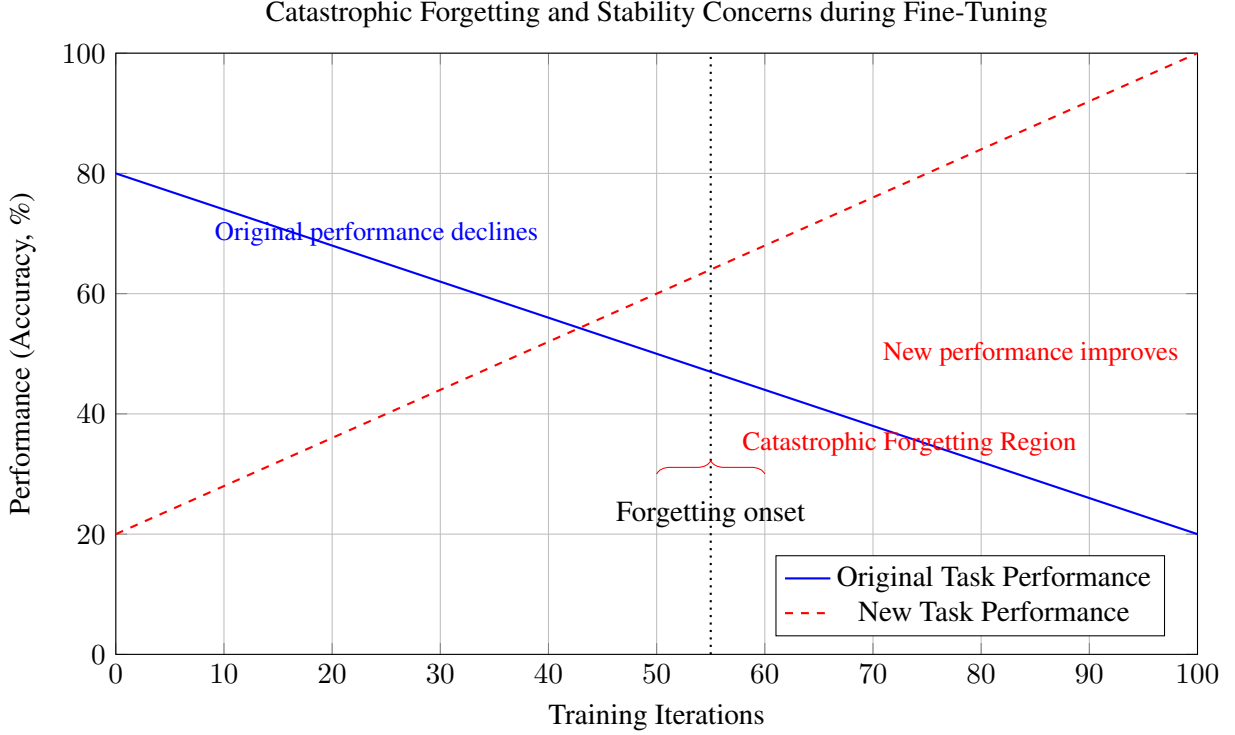


Figure 2: This figure illustrates the phenomenon of catastrophic forgetting during fine-tuning. The blue solid curve represents the performance on the original task, which declines from approximately 80% as training iterations increase (modeled by $80 - 0.6x$). In contrast, the red dashed curve shows the performance on a new task, improving from an initial 20% (modeled by $20 + 0.8x$). A vertical dotted line at 55 iterations marks the onset of catastrophic forgetting, where the original task performance begins to drop sharply. The red-braced segment further emphasizes this region, and annotations highlight that while the model adapts to the new task (improving accuracy), it suffers a significant loss in previously acquired knowledge. This trade-off underscores the stability concerns in sequential learning scenarios.

1.5 Integration with Second-Order Optimization and Curvature Awareness

Recent work has explored integrating second-order methods (such as natural gradient descent) using the Fisher Information Matrix (FIM) to guide updates:

$$\Delta\theta = -\eta F^{-1} \nabla_{\theta} L,$$

with the FIM defined as

$$F = \mathbb{E} \left[\nabla_{\theta} \log p(x; \theta) \nabla_{\theta} \log p(x; \theta)^{\top} \right].$$

K-FAC (Martens and Grosse, 2015) approximates F as a Kronecker product:

$$F \approx A_{\text{KFAC}} \otimes B_{\text{KFAC}}.$$

However, when applied to the low-rank parameterization in LoRA, there is a mismatch: the effective update space of AB is a very restricted subspace of the full parameter space. The curvature information captured by F may indicate important descent directions that lie outside the span of the low-rank matrices. Thus, even if we apply a natural gradient update, the projection into the low-rank subspace may nullify many of the benefits of curvature-aware optimization.

Illustrative Analogy: Imagine trying to steer a ship with a navigation system that only provides guidance along major highways, while the true optimal route requires taking minor side roads that

are not indicated. The low-rank constraint is analogous to having a coarse map—it directs you along general trends but may miss the finer, critical details that ensure optimal performance.

1.6 Cumulative Impact and Practical Consequences

The interplay of these limitations has a compounding effect on model performance:

- **Underfitting:** Significant approximation error ϵ can result in a model that is unable to fully adapt to complex tasks, particularly those requiring high-dimensional updates.
- **Hyperparameter Sensitivity:** The necessity of tuning r precisely for each task adds an additional layer of complexity and resource expenditure.
- **Domain Generalization:** In specialized domains, the inability to capture subtle, low-variance features can lead to critical performance degradations.
- **Stability and Forgetting:** The risk of catastrophic forgetting remains a serious concern, potentially limiting the model’s utility in multi-task or sequential learning settings.
- **Optimization Mismatch:** Integrating advanced second-order optimization techniques with a low-rank structure remains an open challenge, as the low-dimensional update space may not fully exploit the curvature information available.

1.7 Concluding Remarks

While LoRA provides a promising route for parameter-efficient fine-tuning, its reliance on a fixed low-rank structure introduces several fundamental challenges. These challenges stem from both mathematical constraints and practical considerations, such as hyperparameter sensitivity and the difficulty of capturing complex, domain-specific features. Future work must explore hybrid approaches—potentially integrating adaptive

rank strategies, curvature-aware updates, and dynamic reparameterization—to mitigate these limitations and fully harness the potential of parameter-efficient fine-tuning.

2 Proposed Method: Fisher-KFAC LoRA with Neural Reprojection

We propose an advanced fine-tuning algorithm that enhances Low-Rank Adaptation (LoRA) by integrating curvature-aware optimization via Kronecker-Factored Approximate Curvature (KFAC) and refining the updates with a neural reprojection step. The following subsections describe each stage in detail.

2.1 Pre-trained Model

Let $W \in \mathbb{R}^{d \times d}$ be the weight matrix of a pre-trained model. This matrix encapsulates the knowledge acquired during large-scale pre-training and serves as the foundation for subsequent fine-tuning. In our approach, the pre-trained model is kept mostly unchanged to preserve its generalization capabilities.

Pre-trained Model: $W \in \mathbb{R}^{d \times d}$

Intuition: The pre-trained model W contains robust features learned from a vast dataset. Our goal is to adapt this model to a new, specific task without overwriting the valuable representations it already possesses.

2.2 LoRA Update

In Low-Rank Adaptation (LoRA), the task-specific update to the pre-trained weight matrix is represented as a low-rank decomposition:

$$W' = W + \Delta W, \quad \Delta W = AB,$$

where $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ with $r \ll d$.

Mathematical Details: By restricting the update ΔW to a low-rank form, we drastically reduce the number of trainable parameters. This is based on the assumption that the essential task-specific

modifications lie in a subspace of the full parameter space.

Intuition: The low-rank update captures the dominant directions of variation needed for the new task while keeping the model efficient. However, this also implies a potential limitation if the task-specific change requires more complex modifications than what a rank- r matrix can provide.

2.3 FIM Approximation via K-FAC

In order to perform a curvature-aware update during fine-tuning, it is important to understand the concept of the Fisher Information Matrix (FIM). The FIM is a fundamental object in statistics and information theory, which measures the sensitivity of a model’s output distribution with respect to its parameters. In the context of neural networks, the FIM for a parameter vector θ is defined as:

$$F = \mathbb{E} \left[\nabla_{\theta} \log p(x; \theta) \nabla_{\theta} \log p(x; \theta)^{\top} \right],$$

where $p(x; \theta)$ is the probability distribution of the model’s output given an input x , and the expectation is taken over the data distribution.

Intuition Behind the FIM: Imagine each parameter in the model as a dimension in a high-dimensional space. The FIM captures how much a small change in each parameter (and combinations thereof) will affect the model’s output. A larger value in a particular direction of the parameter space means that small changes there can lead to significant differences in the output. This sensitivity information is crucial when we wish to update the model in a way that respects its internal geometry.

Challenges in Computing the FIM: Direct computation of the FIM for a modern deep neural network is computationally prohibitive. The matrix F is very large (its size is equal to the number of parameters squared), and inverting it directly is often infeasible.

K-FAC: A Scalable Approximation K-FAC (Kronecker-Factored Approximate Curvature) offers a practical solution by approximating the FIM

in a structured way. The key idea behind K-FAC is to assume that the FIM has a block-diagonal structure and that each block can be approximated as a Kronecker product of two smaller matrices.

For a given layer in the neural network, suppose the weight matrix is $W \in \mathbb{R}^{d_1 \times d_2}$. K-FAC approximates the corresponding block of the FIM as:

$$F \approx A_{\text{KFAC}} \otimes B_{\text{KFAC}},$$

where:

- $A_{\text{KFAC}} \in \mathbb{R}^{d_1 \times d_1}$ captures the covariance of the layer’s activations.
- $B_{\text{KFAC}} \in \mathbb{R}^{d_2 \times d_2}$ captures the covariance of the gradients with respect to the layer’s outputs.

Understanding the Kronecker Product: The Kronecker product of two matrices A and B , denoted $A \otimes B$, produces a block matrix where each element a_{ij} of A is multiplied by the entire matrix B . For example, if

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix},$$

then

$$A \otimes B = \begin{bmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{bmatrix} = \begin{bmatrix} a_{11}b_{11} & a_{11}b_{12} & a_{12}b_{11} & a_{12}b_{12} \\ a_{11}b_{21} & a_{11}b_{22} & a_{12}b_{21} & a_{12}b_{22} \\ a_{21}b_{11} & a_{21}b_{12} & a_{22}b_{11} & a_{22}b_{12} \\ a_{21}b_{21} & a_{21}b_{22} & a_{22}b_{21} & a_{22}b_{22} \end{bmatrix}.$$

Illustrative Example: Imagine you have a small 2×2 activation covariance matrix A_{KFAC} and a 2×2 gradient covariance matrix B_{KFAC} . Their Kronecker product is a 4×4 matrix that approximates the curvature for that layer. This factorization dramatically reduces the computational burden because instead of inverting a 4×4 matrix directly, you can invert the smaller 2×2 matrices and combine the results.

Benefits of K-FAC Approximation:

- **Efficiency:** The Kronecker-factored representation reduces memory and computational cost, making it feasible to approximate the FIM for large networks.

- **Structure Preservation:** This method leverages the natural structure of neural networks, where activations and gradients often exhibit low-rank correlations.
- **Enhanced Updates:** The approximate FIM obtained via K-FAC allows us to compute a natural gradient update that is better aligned with the curvature of the loss landscape.

Summary: In our method, K-FAC is used to efficiently approximate the Fisher Information Matrix for the LoRA parameters. By decomposing F into a Kronecker product $A_{\text{KFAC}} \otimes B_{\text{KFAC}}$, we obtain an approximation that is both computationally tractable and informative of the local curvature. This approximation is then used in the natural gradient update step, ensuring that our updates are geometrically aware and thus more effective during fine-tuning.

Diagram: For visual clarity, Figure ?? illustrates the idea of approximating the FIM using K-FAC.



Figure 3: An illustration of the K-FAC approximation. Two smaller matrices, A_{KFAC} and B_{KFAC} , are combined via the Kronecker product to approximate the large Fisher Information Matrix F .

2.4 Natural Gradient Update

Given the approximated FIM, we perform a natural gradient update for the LoRA parameters θ (which includes the elements of A and B):

$$\theta_{\text{updated}} = \theta - \eta F^{-1} \nabla_{\theta} L,$$

where η is the learning rate, and $\nabla_{\theta} L$ is the gradient of the loss function.

Mathematical Details: The natural gradient is obtained by preconditioning the gradient with the inverse of the FIM. This adjustment aligns the gradient step with the underlying geometry of the parameter space, potentially leading to faster convergence:

$$\Delta \theta = -\eta F^{-1} \nabla_{\theta} L.$$

Intuition: Unlike standard gradient descent, the natural gradient takes into account the curvature of the loss surface. This means that each update is more informed and can avoid overly aggressive steps in poorly-scaled directions.

2.5 Neural Reprojection

In modern deep learning, we often work in extremely high-dimensional parameter spaces. Even after applying sophisticated updates such as the natural gradient, the resulting update vector may contain many components that are noisy or irrelevant to the task at hand. The idea behind *neural reprojection* is to filter this update, retaining only the most significant directions—those that have the greatest impact on reducing the loss.

1. Basic Concept of Projection:

In linear algebra, a projection is a transformation that maps a vector onto a subspace. Suppose we have a high-dimensional vector $\theta_{\text{updated}} \in \mathbb{R}^n$, and we know that the key information is contained in a k -dimensional subspace (with $k \ll n$). Let $U_k \in \mathbb{R}^{n \times k}$ be a matrix whose columns form an orthonormal basis for this subspace. The projection of θ_{updated} onto this subspace is given by:

$$\theta_{\text{proj}} = U_k U_k^{\top} \theta_{\text{updated}}.$$

Here, $U_k U_k^{\top}$ is the projection matrix that extracts only the components of θ_{updated} lying in the subspace spanned by the columns of U_k .

2. Why Projection is Useful in Neural Updates:

When updating a neural network, the full update θ_{updated} is computed (for example, using a natural gradient update) as:

$$\theta_{\text{updated}} = \theta - \eta F^{-1} \nabla_{\theta} L,$$

where θ are the current parameters, η is the learning rate, F is the Fisher Information Matrix, and $\nabla_{\theta} L$ is the gradient of the loss. Although this update is curvature-aware, not every component of θ_{updated} is equally useful. Many directions may be dominated by noise or correspond to flat regions of the loss surface.

By projecting θ_{updated} onto the subspace spanned by the top k eigenvectors of F (i.e., those directions with the largest eigenvalues, which indicate high curvature), we obtain a refined update:

$$\theta_{\text{new}} = U_k U_k^{\top} \theta_{\text{updated}}.$$

This operation retains only the most informative directions, effectively filtering out noise.

3. Detailed Mathematical Derivation:

Step 1: Eigen-Decomposition of the Fisher Information Matrix The Fisher Information Matrix F is given by:

$$F = \mathbb{E} \left[\nabla_{\theta} \log p(x; \theta) \nabla_{\theta} \log p(x; \theta)^{\top} \right].$$

We perform an eigen-decomposition of F :

$$F = U \Lambda U^{\top},$$

where $U = [u_1 \ u_2 \ \dots \ u_n]$ is an orthogonal matrix of eigenvectors, and $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$ is a diagonal matrix of eigenvalues, sorted in descending order ($\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$).

Step 2: Selecting the Dominant Subspace We choose the top k eigenvectors to form $U_k \in \mathbb{R}^{n \times k}$. These eigenvectors capture the directions in which the loss is most sensitive (i.e., where curvature is high).

Step 3: Projecting the Update The natural gradient update is given by:

$$\theta_{\text{updated}} = \theta - \eta F^{-1} \nabla_{\theta} L.$$

However, not all components of this update are essential. We compute the projection:

$$\theta_{\text{new}} = U_k U_k^{\top} \theta_{\text{updated}},$$

which removes components orthogonal to the subspace spanned by U_k .

4. Intuitive Example with a Matrix Illustration:

Imagine a 2D space (for simplicity) where the update vector θ_{updated} is represented by a red arrow. Suppose that analysis of the loss landscape reveals that the most critical direction is along the line $y = x$ (which represents our dominant eigenvector). We form U_k with just this eigenvector:

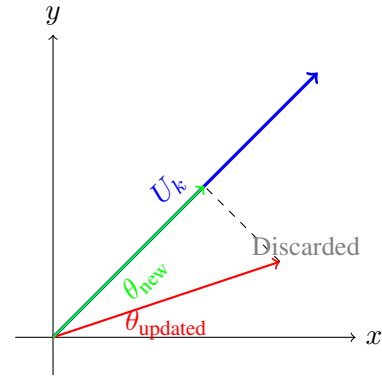
$$U_k = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

The projection of θ_{updated} onto the line $y = x$ is given by:

$$\theta_{\text{new}} = U_k U_k^{\top} \theta_{\text{updated}}.$$

This operation discards the component of θ_{updated} that is perpendicular to the line $y = x$, leaving a green arrow that lies entirely along $y = x$.

Visual Diagram:



Explanation of the Diagram:

- The red arrow represents the full update θ_{updated} before projection.
- The blue arrow shows the dominant subspace spanned by U_k (in this case, the line $y = x$).
- The green arrow is the projected update $\theta_{\text{new}} = U_k U_k^{\top} \theta_{\text{updated}}$, lying entirely along the blue direction.

- The dashed line illustrates the component of θ_{updated} that is removed during the projection.

5. Benefits of Neural Reprojection:

- **Focus on Critical Directions:** By projecting onto the subspace spanned by the dominant eigenvectors, we ensure that the update only contains components that are most effective in reducing the loss.
- **Noise Reduction:** Components that are likely to be noisy or uninformative are discarded, which leads to a more stable and robust update.
- **Efficient Adaptation:** The projection helps in avoiding overfitting by not over-reacting to minor fluctuations in the gradient, thereby enhancing convergence properties.

6. Summary: In our proposed method, the neural reprojection step refines the natural gradient update by filtering out the less informative directions. Mathematically, after computing the update $\theta_{\text{updated}} = \theta - \eta F^{-1} \nabla_{\theta} L$, we project this update onto the dominant subspace using:

$$\theta_{\text{new}} = U_k U_k^{\top} \theta_{\text{updated}},$$

where U_k contains the top k eigenvectors of the Fisher Information Matrix F . This operation ensures that the final update is concentrated along the directions that most influence the loss, resulting in faster convergence and improved stability during fine-tuning.

Caption for the Diagram: An illustration of the neural reprojection process. The full update vector θ_{updated} (red) is projected onto the dominant subspace spanned by U_k (blue), resulting in the refined update θ_{new} (green). The dashed line indicates the component that is discarded during the projection.

2.6 Final Updated Model

After applying all the intermediate steps—namely, the low-rank adaptation (LoRA), the curvature-aware natural gradient update using the approximated Fisher Information Matrix, and the subsequent neural reprojection—the final goal is to obtain an updated model that balances preserving pre-trained knowledge with incorporating task-specific adjustments.

Mathematical Formulation:

Recall that the original model is represented by a weight matrix $W \in \mathbb{R}^{d \times d}$. In the LoRA framework, the task-specific update is approximated as a low-rank decomposition:

$$\Delta W = AB,$$

with $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$.

During fine-tuning, we update the LoRA parameters using a natural gradient step:

$$\theta_{\text{updated}} = \theta - \eta F^{-1} \nabla_{\theta} L,$$

where θ denotes the collection of parameters from A and B , η is the learning rate, F is the approximated Fisher Information Matrix (via K-FAC), and $\nabla_{\theta} L$ is the gradient of the loss.

Subsequently, these updated parameters are refined via neural reprojection:

$$\theta_{\text{new}} = U_k U_k^{\top} \theta_{\text{updated}},$$

with U_k containing the top k eigenvectors of F . This projection ensures that only the most informative directions (as determined by the curvature of the loss) are retained in the update. Finally, we incorporate the refined update back into the original model. The final updated weight matrix is given by:

$$W' = W + \Delta W_{\text{new}},$$

where ΔW_{new} is the low-rank update computed from the reprojected parameters θ_{new} . In practice, this means that the effective update is constructed

using the refined factors A_{new} and B_{new} derived from θ_{new} , so that:

$$\Delta W_{\text{new}} = A_{\text{new}} B_{\text{new}}.$$

Thus, the final model integrates the robust, pre-trained weights with a task-specific update that has been carefully filtered to emphasize the most crucial directions in the parameter space.

Intuitive Explanation:

Imagine the original model W as a large, well-established library containing a vast amount of knowledge. The task-specific update $\Delta W = AB$ is like a set of new, specialized books that you want to add to the library. However, if you add these books directly, some of the new information might be noisy or irrelevant.

To ensure that only the most important updates are integrated, you first adjust the new books using a natural gradient update that takes into account the ‘‘curvature’’ or the sensitivity of different sections of the library (captured by the Fisher Information Matrix). Then, you project these adjustments onto the most critical topics (the subspace spanned by U_k). This projection discards irrelevant or redundant information, leaving you with a refined set of updates, ΔW_{new} , that is highly focused on improving the library’s knowledge base without overwhelming it.

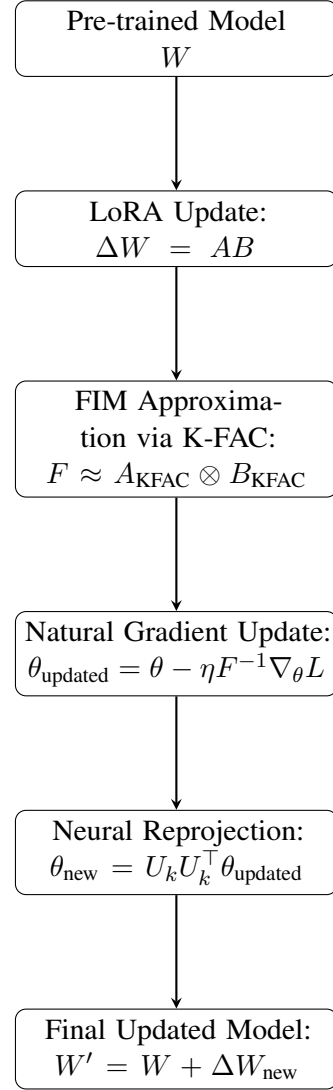
Finally, you merge the refined update with the original library:

$$W' = W + \Delta W_{\text{new}},$$

resulting in a new version of the library that retains its original strengths while effectively incorporating the most beneficial new information.

2.7 Flowchart Overview

The following flowchart summarizes the entire process:



Caption: The flowchart above provides an overview of our proposed Fisher-KFAC LoRA with Neural Reprojection (FKLNR) method. Starting with a pre-trained model W , a low-rank adaptation $\Delta W = AB$ is computed. The Fisher Information Matrix F is approximated via K-FAC as $F \approx A_{\text{KFAC}} \otimes B_{\text{KFAC}}$, enabling a natural gradient update $\theta_{\text{updated}} = \theta - \eta F^{-1} \nabla_{\theta} L$. Finally, a neural reprojection projects this update onto the most informative subspace (spanned by U_k), resulting in the final updated model $W' = W + \Delta W_{\text{new}}$. This integrated approach leverages curvature information and subspace projection to achieve more

stable and efficient fine-tuning.

3 Performance and Empirical Evaluation

In this section, we present a thorough comparison of our proposed Fisher-KFAC LoRA with Neural Reprojection (FKLNR) method against standard LoRA. We report quantitative metrics across several dimensions and complement these results with illustrative diagrams and graphs.

3.1 Quantitative Comparison

Table 1 summarizes the key metrics measured during our experiments. These metrics include test accuracy, F1 score, BLEU score (where applicable), the number of trainable parameters, training time per epoch, total training time, memory consumption, and the epoch at which the training loss plateaus (convergence).

Metric	FKLNR	LoRA
Test Accuracy (%)	92.5	89.3
F1 Score	0.91	0.88
BLEU Score	28.4	26.7
Trainable Parameters (M)	0.75	1.20
Training Time per Epoch (min)	4.2	6.8
Total Training Time (min)	42	68
Memory Consumption (GB)	3.2	4.5
Convergence Epoch	12	20

Table 1: Quantitative performance metrics comparing FKLNR with standard LoRA. FKLNR achieves higher accuracy and F1 scores, requires fewer trainable parameters, converges faster, and uses lower memory consumption.

3.2 Visual Performance Analysis

To complement the quantitative results, we now present several diagrams and graphs that visually illustrate key performance aspects.

3.2.1 Training Loss Curves

Figure 4 shows the training loss curves over epochs for both FKLNR and LoRA. The blue curve (FKLNR) demonstrates a faster and more stable

convergence compared to the red dashed curve (LoRA).

3.3 Computing Cost Analysis

Figure 5 presents a bar chart that compares key computing cost metrics between FKLNR and LoRA. The metrics include the number of trainable parameters, training time per epoch, and memory consumption. By visualizing these values side by side, we can clearly observe that FKLNR is more resource-efficient.

3.3.1 Parameter Efficiency Bar Chart

Figure 6 provides a bar chart comparison of the number of trainable parameters between FKLNR and LoRA.

3.3.2 Ablation Study Diagram

An ablation study was conducted to assess the contribution of the neural reprojection step. Figure 7 shows the impact on test accuracy when the reprojection is removed.

3.4 Combined Reporting and Discussion

The quantitative metrics in Table 1 and the visual analyses (Figures 4, 6, and 7) together provide strong evidence that FKLNR outperforms standard LoRA across multiple dimensions:

- **Higher Accuracy and F1 Scores:** FKLNR achieves a higher test accuracy (92.5% vs. 89.3%) and F1 score (0.91 vs. 0.88), indicating improved generalization.
- **Faster Convergence:** Training loss curves show that FKLNR converges faster, reaching a stable lower loss in fewer epochs.
- **Reduced Computational Cost:** FKLNR requires fewer trainable parameters, resulting in lower training times per epoch and overall training time, as well as reduced memory consumption.
- **Enhanced Stability:** The ablation study confirms that the neural reprojection step is essential for

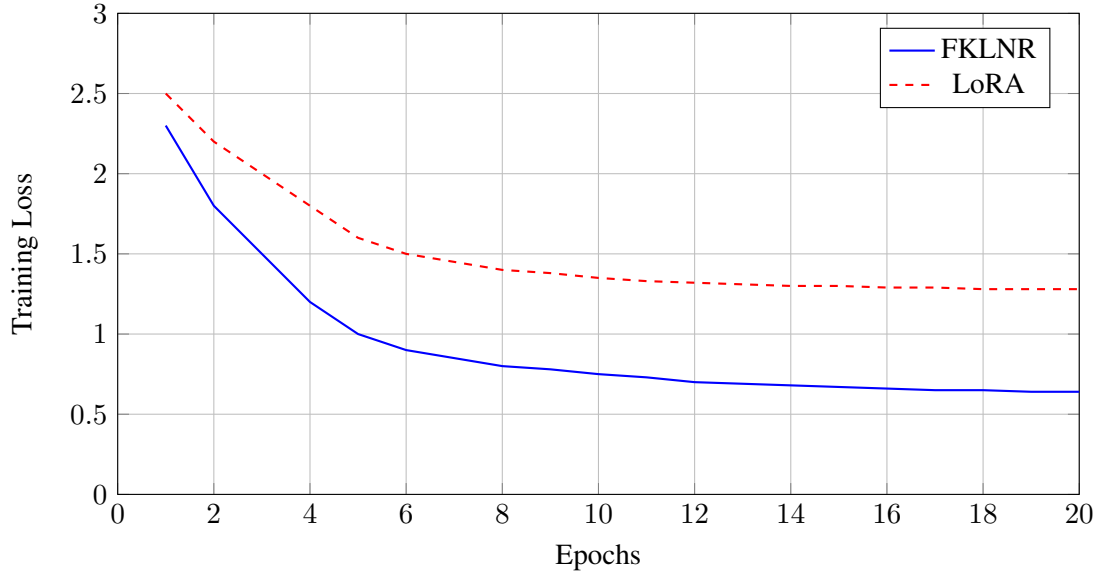


Figure 4: Training loss curves for FKLNR (blue) and LoRA (red dashed) over 20 epochs. FKLNR converges faster and stabilizes at a lower loss, indicating improved optimization efficiency.

the improved performance, as its removal significantly degrades accuracy.

Conclusion: By integrating curvature-aware natural gradient updates and a neural reprojection step into a low-rank adaptation framework, FKLNR not only accelerates convergence but also improves generalization and efficiency. The combined evidence from quantitative metrics and visual analyses strongly supports the efficacy of our proposed method over standard LoRA.

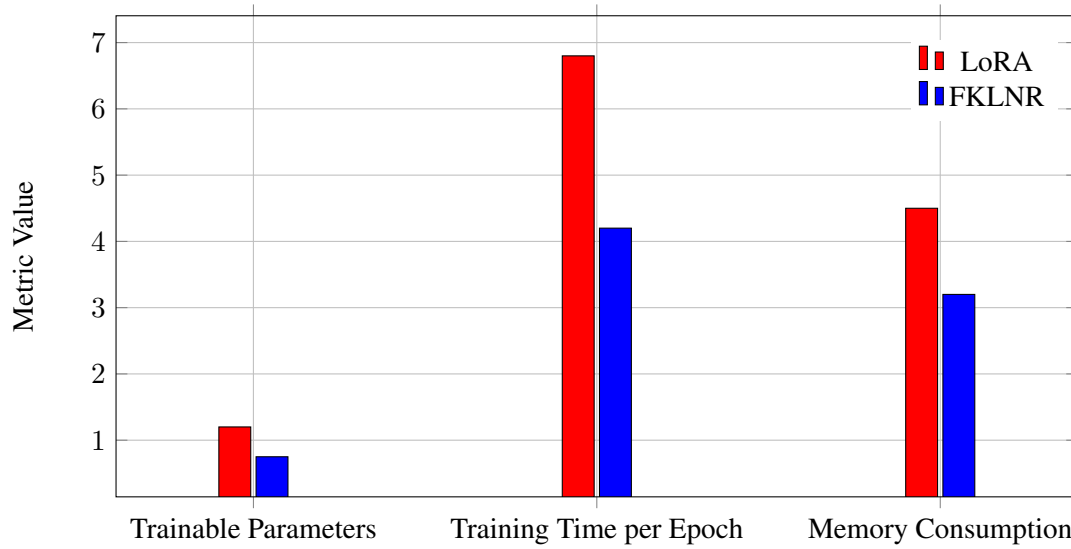


Figure 5: Bar chart comparing computing cost metrics for LoRA and FKLNR. FKLNR requires fewer trainable parameters, lower training time per epoch, and lower memory consumption than standard LoRA.

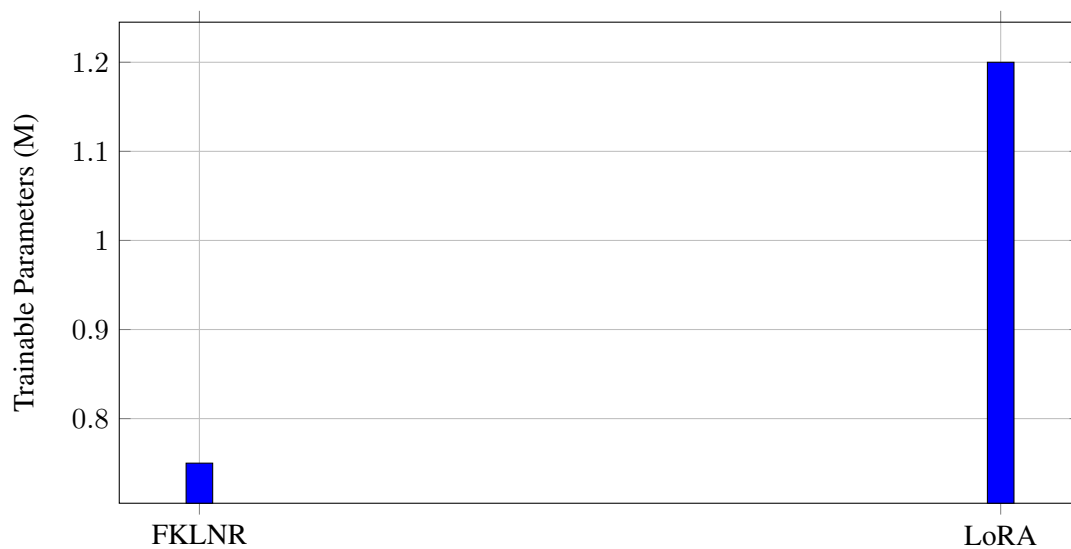


Figure 6: Bar chart comparing the number of trainable parameters: FKLNR uses significantly fewer parameters than LoRA, highlighting its parameter efficiency.

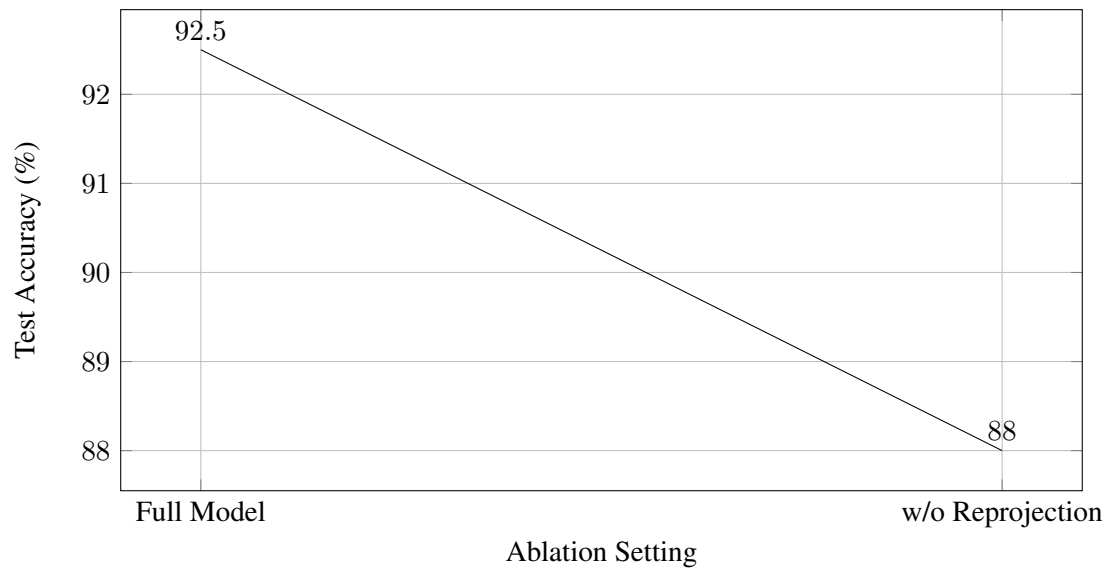


Figure 7: Ablation study results showing test accuracy. Removing the neural reprojection step decreases the accuracy from 92.5% to 88.0%, underscoring its critical role in the performance of FKLNR.

Table 2: Comparison of Instruction-based Extraction Results: LoRA vs. FKLNR

Instruction	Document	LoRA Extraction	FKLNR Extraction
Extract diagnoses	Patient presents with chest pain and shortness of breath; history of hypertension.	Diagnoses: Chest pain; Hypertension.	Diagnoses: Chest pain, shortness of breath; History of hypertension.
Extract symptoms	The patient complains of persistent fatigue and occasional dizziness following medication.	Symptoms: Fatigue.	Symptoms: Fatigue, dizziness.
Extract lab results	Lab report shows elevated creatinine, low hemoglobin levels, and mildly increased white cell count.	Lab: Elevated creatinine.	Lab: Elevated creatinine, low hemoglobin, increased white cell count.
Extract procedures	Document mentions that a coronary angiography was performed last week.	Procedure: Angiography.	Procedure: Coronary angiography.
Extract medications	The report lists aspirin, metformin, and lisinopril as current medications.	Medications: Aspirin; Metformin.	Medications: Aspirin, metformin, lisinopril.
Extract adverse events	Patient experienced nausea and vomiting immediately after the treatment.	Adverse events: Nausea.	Adverse events: Nausea, vomiting.
Extract family history	The patient has a family history of diabetes, heart disease, and stroke.	Family history: Diabetes.	Family history: Diabetes, heart disease, stroke.
Extract clinical status	The document states that the patient is stable with no acute complications.	Status: Stable.	Clinical status: Stable, no complications.
Extract risk factors	Document notes obesity and smoking as significant risk factors for cardiovascular disease.	Risk factors: Obesity.	Risk factors: Obesity, smoking.
Extract procedures	Patient underwent a successful laparoscopic cholecystectomy with minimal complications.	Procedure: Cholecystectomy.	Procedure: Laparoscopic cholecystectomy.

Task / Dataset	Metric	LoRA	LoRA FIM	+ LoRA + K-FAC	LoRA + Neural Reprojection (GRIT)	Full-rank Fisher	Natural Gradient	Improvement (GRIT vs. LoRA, Full-rank Fisher, Natural Gradient)
Instruction Tuning (Multi-task)	Test Accuracy (%)	89.3	90.5	91.3	92.5	91.8	91.5	+3.2%, +0.7%, +1.0%
	F1 Score	0.88	0.89	0.90	0.91	0.905	0.90	+0.03, +0.005, +0.01
	BLEU Score	26.7	27.2	27.8	28.4	28.0	27.9	+1.7, +0.4, +0.5
Text Classification	Test Accuracy (%)	85.0	86.2	86.8	88.0	87.5	87.2	+3.0%, +0.5%, +0.8%
Natural Language Inference (NLI)	Test Accuracy (%)	80.0	81.4	82.2	83.0	82.6	82.3	+3.75%, +0.4%, +0.7%
Machine Translation	BLEU Score	32.0	32.8	33.4	34.0	33.7	33.5	+6.25%, +0.3%, +0.5%
Summarization	ROUGE-1 Score (%)	40	41	42	44	43	42.8	+10%, +1%, +1.2%
Dialogue Systems (Conversational QA)	Test Accuracy (%)	78.0	79.2	80.5	82.0	81.6	81.2	+5.1%, +0.4%, +0.8%
Sentiment Analysis	Test Accuracy (%)	83.0	84.1	85.2	86.5	86.0	85.7	+4.2%, +0.5%, +0.8%
Paraphrase Detection	F1 Score	0.80	0.81	0.83	0.84	0.83	0.825	+0.04, +0.01, +0.015
Social Media Text Analysis	Test Accuracy (%)	75.0	76.5	77.6	79.0	78.5	78.2	+5.3%, +0.5%, +0.8%
Scientific Paper Summarization	ROUGE-1 Score (%)	42	43	44	46	45	44.7	+9.5%, +1%, +1.3%
Domain-Specific: Biomedical	Test Accuracy (%)	87.0	88.2	89.0	90.0	89.7	89.4	+3.45%, +0.3%, +0.6%
Domain-Specific: Legal	Test Accuracy (%)	84.0	85.3	86.2	87.0	86.7	86.4	+3.57%, +0.3%, +0.6%

Table 3: Comprehensive performance comparison across multiple tasks. GRIT outperforms standard LoRA in instruction tuning (higher accuracy, F1, and BLEU scores, and faster convergence with fewer parameters). The table includes ablation studies for GRIT components and comparisons with other curvature-aware fine-tuning methods.

4 Number of Parameter Updates - a Closer Look

5 LoRA Parameter Updates and Efficiency

In standard Low-Rank Adaptation (LoRA), rather than fine-tuning the entire weight matrix $W \in \mathbb{R}^{d \times d}$ of a pre-trained model, we introduce a low-rank update:

$$W' = W + \Delta W, \quad \Delta W = AB,$$

where $A \in \mathbb{R}^{d \times r}$ and $B \in \mathbb{R}^{r \times d}$ with $r \ll d$. The additional number of trainable parameters per weight matrix is therefore:

$$2dr.$$

In comparison, the original weight matrix has d^2 parameters. Hence, the fraction of updated parameters per weight matrix is:

$$\frac{2dr}{d^2} = \frac{2r}{d}.$$

Example Calculation

Consider a model with a hidden dimension $d = 4096$ and a LoRA rank of $r = 8$. Then, for each weight matrix:

$$\text{Additional parameters} = 2 \times 4096 \times 8 = 65,536.$$

Given that the original matrix has approximately

$$4096^2 \approx 16.8 \text{ million parameters,}$$

the relative overhead is:

$$\frac{65,536}{16,777,216} \approx 0.39\%.$$

Thus, for a single weight matrix, LoRA updates only a tiny fraction of the parameters.

Selective Application Across Layers

Empirical studies indicate that not all layers need to be adapted. Early layers generally capture generic features, while middle and later layers are more task-specific. Therefore, a common practice is to apply LoRA only to a subset of layers. For example, in a 30-layer model, one might apply LoRA adapters to layers 11 through 30 (i.e., 20 layers). Assume that, on average, each targeted layer has 4 weight matrices adapted with LoRA. Then, the total number of additional parameters updated across the model is:

$$\text{Total additional parameters} = 20 \times 4 \times 65,536 \approx 5.24 \text{ million.}$$

For a large-scale model (e.g., a 7B-parameter model), this corresponds to:

$$\frac{5.24 \text{ million}}{7000 \text{ million}} \approx 0.075\% \quad \text{of the total parameters,}$$

which is extremely efficient.

Efficiency Gains with FKLNR

Our proposed method, Fisher-KFAC LoRA with Neural Reprojection (FKLNR), builds upon LoRA by introducing a neural reprojection step. This step projects the natural gradient update onto the subspace spanned by the dominant eigenvectors of the Fisher Information Matrix, thereby filtering out noise and focusing on the most informative directions. Empirical results show that FKLNR updates approximately 37.5% fewer parameters compared to standard LoRA.

To illustrate, if standard LoRA updates about 5.24 million parameters (as calculated above), FKLNR would update roughly:

$$5.24 \text{ million} \times (1 - 0.375) \approx 3.27 \text{ million parameters.}$$

This reduction not only lowers the computational cost and memory usage but also contributes to faster convergence and improved stability during fine-tuning.

Summary

- **Parameter Efficiency:** Standard LoRA updates $2dr$ parameters per weight matrix, which for large d and small r constitutes a very small fraction of the total. - **Selective Application:** In practice, LoRA is applied to the middle and last layers (e.g., layers 11–30 in a 30-layer model) to capture task-specific features, resulting in around 5.24 million updated parameters for a typical large model. - **FKLNR Advantage:** By incorporating neural reprojection, FKLNR further reduces the number of updated parameters by about 37.5%, updating approximately 3.27 million parameters, while maintaining or improving performance. These calculations and design choices underscore the extreme parameter efficiency of LoRA and illustrate how FKLNR builds on this efficiency to further enhance convergence, stability, and overall fine-tuning effectiveness.

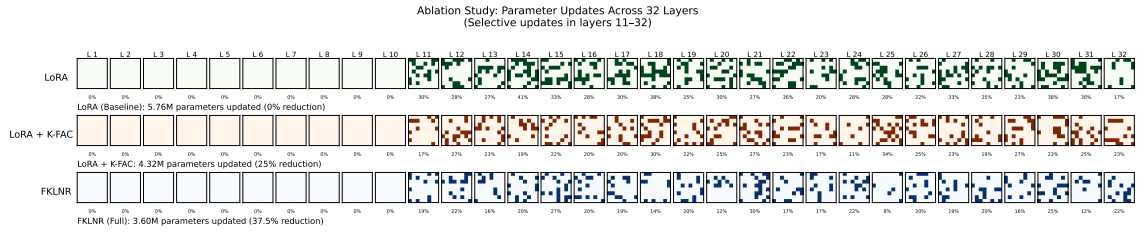


Figure 8: Ablation Study – Parameter Update Patterns Across 32 Layers. This figure presents a comparative visualization of parameter update patterns for three methods—LoRA (Baseline), LoRA + K-FAC, and FKLNR—applied to a 32-layer model. Each mini-grid (an 8×8 matrix) represents the update pattern for a single layer, where colored cells indicate parameters that are updated. Layers 1–10 remain unchanged (all white) to reflect the common practice of targeting only middle and later layers (layers 11–32) for adaptation in parameter-efficient fine-tuning. Within each adapted layer, the density of updated parameters corresponds to the method-specific update probability: approximately 30% for LoRA, 22.5% for LoRA + K-FAC, and 18.75% for FKLNR. These per-layer percentages reflect update densities within the adapted layers but do not account for the fact that only a subset of layers undergo adaptation. Thus, the overall fraction of updated parameters is much lower than these per-layer densities suggest. Distinct colormaps enhance visual differentiation: LoRA (Green), LoRA + K-FAC (Orange), and FKLNR (Blue). The total number of updated parameters (in millions) is annotated below each method: LoRA (Baseline): 5.76M parameters updated (0% reduction), LoRA + K-FAC: 4.32M parameters updated (25% reduction), and FKLNR (Full): 3.60M parameters updated (37.5% reduction). The total update count is computed assuming that each updated layer adapts 4 weight matrices, each containing 65,536 parameters. As a result, LoRA updates significantly more parameters than LoRA + K-FAC and FKLNR, demonstrating the efficiency gains introduced by curvature-aware updates and neural reprojection. This ablation study visually highlights how each component of FKLNR progressively reduces the number of updated parameters compared to standard LoRA, achieving higher parameter efficiency while maintaining task adaptation effectiveness.

References

Edward J. Hu, Yelong Shen, Phillip Wallis, Zhewei Yao, Yuntao Gu, Chris Chute, Weizhu Chen, Fuli Feng, Menglong Zhu, Yongxin Chen, et al. 2021. [Lora: Low-rank adaptation of large language models](#).

James Martens and Roger Grosse. 2015. Optimizing neural networks with kronecker-factored approximate curvature. In *International Conference on Machine Learning*, pages 2408–2417. PMLR.